

AI-Powered Collections Strategy

Geldium Financial Services — Executive Briefing

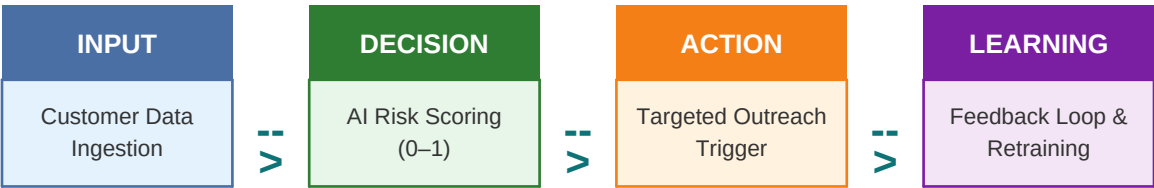
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This deck outlines a high-level concept for an autonomous, responsible AI-powered collections system — translating predictive risk insights into scalable, fair, and explainable customer outreach.

Slide 1–2 | How the AI Collections System Works

System Overview

The AI collections system operates as a continuous four-stage pipeline — ingesting customer data, generating risk scores, triggering personalised interventions, and learning from outcomes to improve over time.



- **Stage 1 — Data Inputs:** Credit utilization rate, payment history, DTI ratio, annual income, open credit lines. Data is cleaned via median/mean imputation before scoring.
- **Stage 2 — Decision Logic (AI Risk Scoring):** Random Forest Classifier generates a delinquency probability score (0–1) for each customer. Score thresholds: **Red (>0.7)** = high risk, **Amber (0.4–0.7)** = moderate, **Green (<0.4)** = low risk.
- **Stage 3 — Actions (Personalised Outreach):** Red: Immediate call from collections officer + repayment plan offer. Amber: Automated SMS/email reminder + financial wellness resources. Green: Standard statement communication — no intervention required.
- **Stage 4 — Learning Loop:** Model retrains monthly on new outcomes (repayment, default, response rate). Performance drift is monitored via AUC-ROC and F1-Score dashboards.

Slide 3 | Role of Agentic AI — Autonomous vs. Human Oversight

The system uses a **human-in-the-loop** design — maximising automation efficiency while ensuring accountability for high-stakes decisions.

Autonomous (AI-Driven)	Human Oversight Required
Risk score calculation for all customers (daily batch)	Review of all Red-flag accounts before any contact
Automated SMS/email reminders for Amber-tier customers	Approval of repayment plan terms and amounts

Real-time dashboard updates and KPI reporting	Handling customer disputes or complaints about AI flags
Routine data preprocessing and feature engineering	Quarterly model fairness audits and bias reviews
Monthly model retraining on new outcome data	Final decision on account escalation or legal action

Slide 4 | Responsible AI Guardrails

The system is designed with four core guardrails to ensure it operates **fairly, transparently, and in compliance** with financial services regulations.

- **Fairness & Bias Monitoring:** Disparate Impact Ratio (DIR) checks run monthly across age, location, and income bands. If any subgroup shows a delinquency flag rate more than 20% higher than the baseline, the model is paused for review.
- **Explainability (XAI):** Every AI flag includes a plain-language reason visible to collections staff — e.g., *"Flagged due to 85% credit utilization and 2 missed payments."* Staff can interrogate any score using SHAP value breakdowns.
- **Regulatory Compliance:** All model decisions are logged with full audit trails. The system complies with RBI fair lending guidelines and GDPR data minimisation principles — no unnecessary personal data is processed.
- **Human Override:** Any customer can request a human review of their AI-generated risk score. Collections officers have override authority at all tiers. No automated action contacts a customer more than once per 48 hours.

Compliance Commitment: All AI predictions serve as decision-support tools only. No credit decision, account restriction, or legal escalation is made by the AI alone. Model governance is reviewed by the Risk & Compliance team quarterly.

Slide 5 | Expected Business Impact

The AI-powered collections system is projected to deliver measurable improvements across both **business performance** and **customer experience** within 12 months of deployment.

Business KPIs (Quantitative)	Customer Outcomes (Qualitative)
15% reduction in 90-day delinquency rate within 6 months	Personalised, empathetic outreach — right message, right time
20–30% decrease in collections operational cost via automation	Reduced financial stress through early intervention and repayment plans
25% improvement in early intervention success rate (Amber tier)	Transparent, explainable decisions customers can understand and challenge
40% reduction in manual triage time for collections officers	Fairer treatment — no demographic group unfairly over-targeted

10% increase in repayment plan uptake through targeted SMS/email

Improved trust in Geldium through responsible, human-reviewed AI

Strategic Value: Beyond cost savings, this system positions Geldium as a **responsible AI leader** in financial services — demonstrating that automation and ethical customer care are not mutually exclusive. Scalability means the same pipeline can be extended to new products and markets with minimal re-engineering.